Hyunsik Jeon

AI Strategist

Portfolio: https://portfolio-app-5ca9.vercel.app/

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FEATURED PROJECT 01

AIIM — AI Influencer Match

Hyunsik Jeon (Solo)

Python · FastAPI · Next.js 14 · LangGraph · GraphRAG · PostgreSQL (Cloud SQL) · Selenium · YouTube Data API · Meta Graph API · LLM · GCP Cloud Run

An end-to-end cloud platform that ingests 5,300+ creators across YouTube + Instagram + TikTok in 14 languages and **ranks brand-fit via a LangGraph agentic workflow layered on GraphRAG retrieval** — returning shortlists with per-pick rationale. Multilingual hashtag discovery surfaced 400 new YouTube creators in one 30-min run; extracted 323 IG business emails. Runs 24/7 on Google Cloud (Cloud Run + Cloud SQL). **Live:** aiim-frontend-559416574965.asia-northeast3.run.app · **GitHub:** github.com/jeonhs9110/AIIM-Showcase

Technical Pipeline Detail

1. LangGraph agentic workflow Brief parsing → candidate retrieval → multi-criteria scoring → rationale writing, modelled as a state graph (with retries, branching, human-in-the-loop hooks)

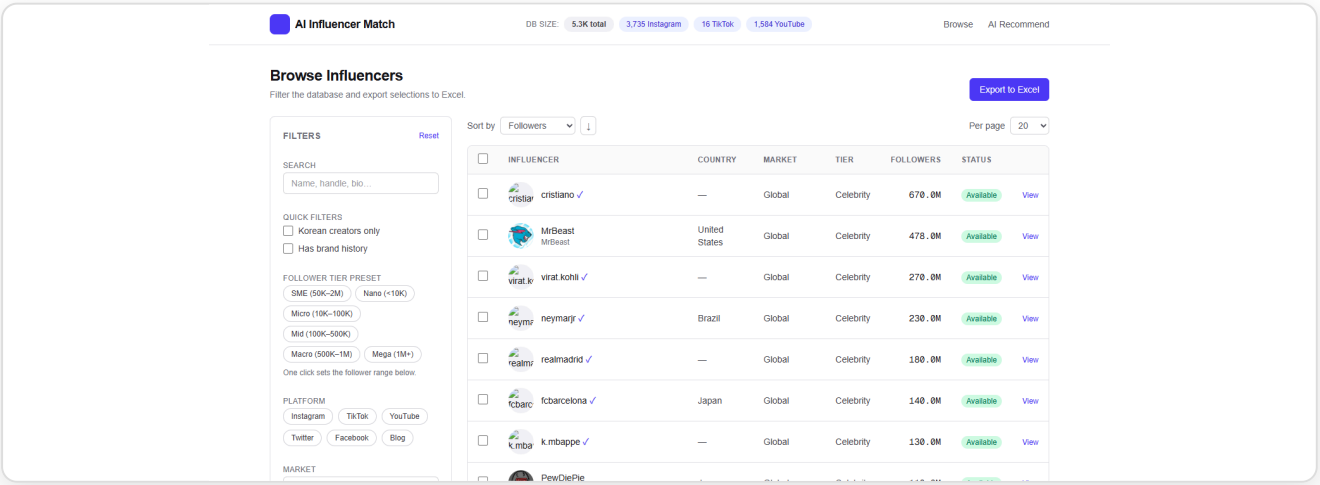
2. GraphRAG retrieval Creator · category · brand-collab relations encoded as a knowledge graph so every pick comes with a structural "why this creator" rationale

3. Multi-platform ingestion YouTube (1,584) · Instagram (3,735) · TikTok (16) — 5,300+ creators total

4. Multilingual hashtag strategy Searches 14 writing systems (광고 · スポンサー · رعاية …) to escape English-query convergence

5. LLM enrichment Country · language · brand-collab history · vertical · contacts — auto-extracted with prompt caching

6. Infra GCP asia-northeast3 — Cloud Run + Cloud SQL Postgres + Cloud Build CI



FEATURED PROJECT 02

KOKKOK Garden — K-Beauty Multi-region E-commerce Hyunsik Jeon (Solo)

Next.js 16 · TypeScript · Supabase (PostgreSQL + Auth + Storage) · Tailwind CSS 4 · OpenAI GPT-4o mini · Vercel

A sustainable e-commerce designed for an administrator to keep running long after the initial build. Six core data entities (products · orders · cart · users · media_stories · YouTube Shorts) are normalised into a Supabase (PostgreSQL) schema under proper RLS and relational constraints; every entity has a unified `/admin` CRUD + image-upload + state flow so a non-developer can operate the store. KR (purchasable) / GL (viewing + AI chatbot) routing and GPT-4o mini 5-language product translation all run from one codebase. **Live:** https://kokv2.vercel.app · **GitHub:** github.com/jeonhs9110/kok_v2

Technical Pipeline Detail

1. Sustainable admin tooling `/admin` dashboard — product · Shorts · media CRUD + image upload, operable without a developer

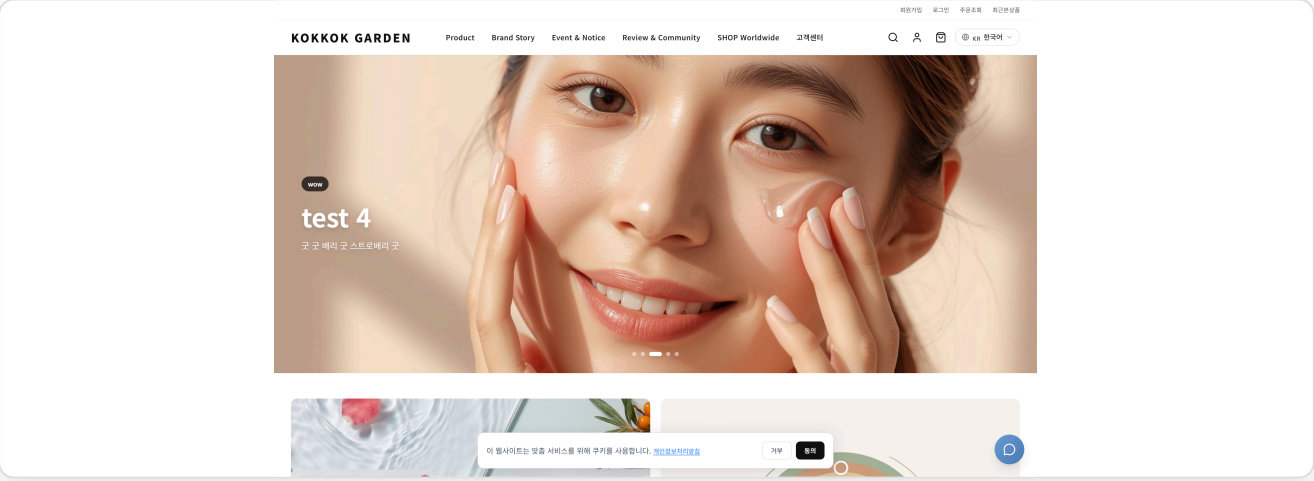
2. Normalised DB schema Six entities on Supabase (PostgreSQL) — users · products · orders · cart_items · media_stories · shorts — with RLS and relational constraints

3. Unified auth Supabase Auth + cookie-based admin session — login, permissions, and RLS handled in one layer

4. Multi-region routing KR/GL split via `x-user-country`, six language routes

5. AI translation + chatbot GPT-4o mini 5-language translation + unstable_cache 24h caching, with a K-beauty consultation chatbot on the GL store

6. Deploy Vercel · https://kokv2.vercel.app



FEATURED PROJECT 03

IAMX6 — Naver Blog Automation Suite

Hyunsik Jeon (Solo · source private)

Electron · Puppeteer-core · Node.js · LLM · Bytenode (V8) · Electron Fuses · Windows portable .exe

Key impact: **a viral-marketing engine for Naver blog owners**. Automates neighbor requests · comment/like · visitor replies with human-shaped behavior to pile up the inbound / dwell / interaction signals the Naver algorithm rewards — deliberately pushing posts toward **top search ranking (검색 상위 노출)**. Distributed as a single ~67MB portable .exe, three-layer hardened (control-flow-flattened JS + V8 bytecode + Electron Fuses). **GitHub:** github.com/jeonhs9110/IAMX6-showcase

> **Technical Pipeline Detail**
>
> **1. Viral-marketing impact** Accumulates inbound / dwell / interaction signals → maximises probability of **top search ranking (검색 상위 노출)** on Naver search and blog rankings
>
> **2. 3-workflow unified GUI** Neighbor-request · comment/like · per-visit reply + settings tab
>
> **3. Puppeteer automation** Drives a real Naver session — 2FA/CAPTCHA handled manually by the user, then the bot takes over
>
> **4. LLM comment generation** Post title + body → context-shaped Korean comments, 6–10 cached per run
>
> **5. Human simulation** Dwell time scaled to character count + ±25% jitter + ms-level noise + 35% "distraction" actions
>
> **6. 3-layer hardening** javascript-obfuscator → bytenode (.jsc) → Electron Fuses — blocks `asar extract`, debugger attach, and run-as-Node

FEATURED PROJECT 04

Top Award — IBM x RedHat AI Transformation Program

Vaccine Daily — AI News Intelligence Pipeline

Hyunsik Jeon and 2 others

React · FastAPI · PostgreSQL · ChromaDB · HDBSCAN · ko-sroberta · Kiwi · KR-WordRank · GPT-4o-mini · GraphRAG · Docker · AWS EC2

Cutting-edge AI for cross-outlet reporting-tone comparison. A news-automation pipeline that systematically cuts LLM API cost by composing multiple Python libraries around GPT-4o-mini, and kills hallucinations with a Writer → Critic → Editor 3-agent system. HDBSCAN + ko-sroberta clustering eliminates duplicate LLM calls, a two-stage Kiwi morphological filter catches most obvious cases before the LLM, and a ChromaDB embedding cache prevents re-encoding. **Live:** jeonhs9110.github.io/vaccine-daily/ · **GitHub:** github.com/jeonhs9110/vaccine-daily

Technical Pipeline Detail

1. Multi-agent system Writer → Critic → Editor — Critic judges the draft against the source as PASS/FAIL for hallucination/fact-distortion, auto-retry on failure, async parallel up to 5 at a time

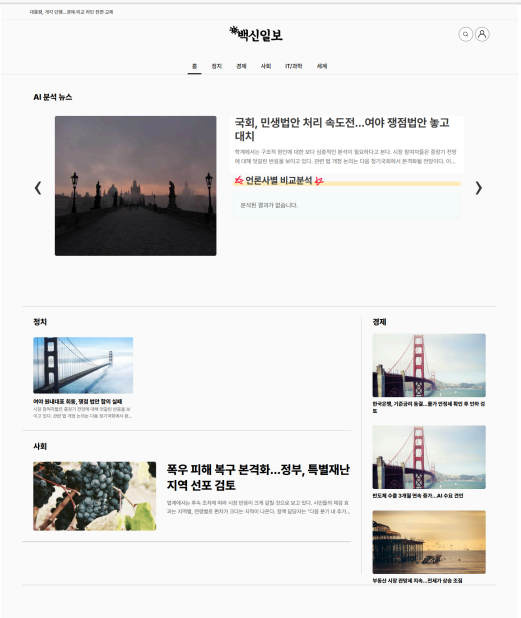
2. LLM cost reduction via multiple Python libraries HDBSCAN clustering + ko-sroberta embeddings + two-stage Kiwi morphological filter + ChromaDB embedding cache — blocks duplicate LLM calls and re-encoding

3. Data collection Selenium + BeautifulSoup crawl 10 media outlets on a 5-minute cycle

4. GraphRAG tone analysis Entity-Relation triples → knowledge graph visualising per-outlet reporting tone and framing; only the graph summary is injected into the prompt

5. Personalised recommendation Subscribed keywords + read history + category preference → multi-criteria scoring + TextRank + KR-WordRank ensemble + D3 word cloud

6. Deploy Docker Compose + Nginx + Certbot (HTTPS) + AWS EC2



FEATURED PROJECT 05

FOBO AI — European Football Prediction

Hyunsik Jeon (Solo)

Python · PyTorch · Transformer · Graph Attention Network · PPO RL · Dixon-Coles · XGBoost · LightGBM · Flask · Google Cloud

A Machine Learning and Reinforcement Learning pipeline for football-match prediction. End-to-end ML system chaining PyTorch Transformer + GAT + Dixon-Coles correction + PPO RL + XGBoost/LightGBM into an 8-stage pipeline. Across 13 European leagues: ~90% hit rate at ensemble ≥77% + favourite filter; ~95% when the PPO agent independently agrees at ≥90%. Deployed on Google Cloud via a 2-VM architecture for ~$26/month. **Live:** http://34.64.147.124/ · **GitHub:** github.com/jeonhs9110/epl-showcase

8-Step System Architecture

1. Scrape Parallel Flashscore crawl across 13 leagues (incremental re-runs)

2. Validation Integrity checks catch broken scrapes before they poison training

3. Seq optimisation Per-league look-back window (Premier ~10, Championship ~5)

4. DL training Transformer + GAT + Dixon-Coles NLL — infers defensive posture in weak-vs-strong matchups

5. PPO RL Action ∈ {Home, Draw, Away, Pass} with Kelly-Criterion-shaped reward

6. Hybrid ensemble XGBoost + LightGBM on DL embeddings

7. Calibration Platt scaling aligns reported probability with empirical hit rate

8. Flask API Models preloaded, <200 ms per request, GPT-4o-mini chatbot scoped to today's matches